5.1 Real Integrals

To The Instructor : We present this section as a review of the definitions and methods of evaluation of the definite integral and line integrals in the plane. In our experience we have found that a re-examination of this material contributes to a smoother introduction to complex integration. You can, of course, skip this section and move directly into complex contour integrals if you think your students have adequate familiarity with these concepts. However, the terminology about curves in the plane introduced in this section will be used in the succeeding section.

INTRODUCTION It is likely that you have retained at least two associations from your study of elementary calculus: the derivative with slope, and the definite integral with area. But as the derivative $f'(x)$ of a real function $y = f(x)$ has other uses besides finding slopes of tangent lines, so too the value of a definite integral $\int_a^b f(x)\,dx$ need not be area "under a curve." Recall, if $F(x)$ is an antiderivative of a continuous function f, that is, F is a function for which $F'(x) = f(x)$, then the definite integral of f on the interval $[a, b]$ is the number

$$\int_a^b f(x)\,dx = F(x)\Big|_a^b = F(b) - F(a). \tag{1}$$

For example, $\int_{-1}^{2} x^2 dx = \frac{1}{3}x^3\Big|_{-1}^{2} = \frac{8}{3} - \left(-\frac{1}{3}\right) = 3$. Bear in mind that the **fundamental theorem of calculus**, just given in (1), is a method of *evaluating* $\int_a^b f(x)\,dx$; it is not the *definition* of $\int_a^b f(x)\,dx$.

Definite Integral

In the discussion that follows we present the definitions of two types of **real integrals**. We begin with the five steps leading to the definition of the definite (or Riemann) integral of a function f; we follow it with the definition of line integrals in the Cartesian plane. Both definitions rest on the limit concept.

Steps Leading to the Definition of the Definite Integral

1. *Let f be a function of a single variable x defined at all points in a closed interval $[a, b]$.*
2. *Let P be a partition:*

$$a = x_0 < x_1 < x_2 < \cdots < x_{n-1} < x_n = b$$

of $[a, b]$ into n subintervals $[x_{k-1},\ x_k]$ of length $\Delta x_k = x_k - x_{k-1}$. See FIGURE 5.1.1.

3. *Let $\|P\|$ be the **norm** of the partition P of $[a, b]$, that is, the length of the longest subinterval.*
4. *Choose a number x_k^* in each subinterval $[x_{k-1},\ x_k]$ of $[a, b]$. See Figure 5.1.1.*
5. *Form n products $f(x_k^*)\Delta x_k$, $k = 1, 2, \ldots, n$, and then sum these products:*

$$\sum_{k=1}^{n} f(x_k^*)\,\Delta x_k.$$

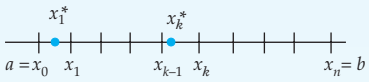


FIGURE 5.1.1 Partition of $[a,\ b]$ with x_k^* in each subinterval $[x_{k-1},\ x_k]$

DEFINITION 5.1.1 Definite Integral

The **definite integral** of f on $[a, b]$ is

$$\int_a^b f(x)\,dx = \lim_{\|P\|\to 0} \sum_{k=1}^{n} f(x_k^*)\,\Delta x_k. \tag{2}$$

Whenever the limit in (2) exists we say that f is **integrable** on the interval $[a, b]$ or that the definite integral of f **exists**. It can be proved that if f is continuous on $[a, b]$, then the integral defined in (2) exists.

The notion of the definite integral $\int_a^b f(x)\,dx$, that is, *integration of a real function $f(x)$ over an interval on the x-axis from $x = a$ to $x = b$* can be generalized to *integration of a real multivariable function $G(x, y)$ on a curve C from point A to point B* in the Cartesian plane. To this end we need to introduce some terminology about curves.

Terminology Suppose a curve C in the plane is parametrized by a set of equations $x = x(t)$, $y = y(t)$, $a \le t \le b$, where $x(t)$ and $y(t)$ are continuous real functions. Let the **initial** and **terminal points** of C, that is, $(x(a), y(a))$ and $(x(b), y(b))$, be denoted by the symbols A and B, respectively. We say that:

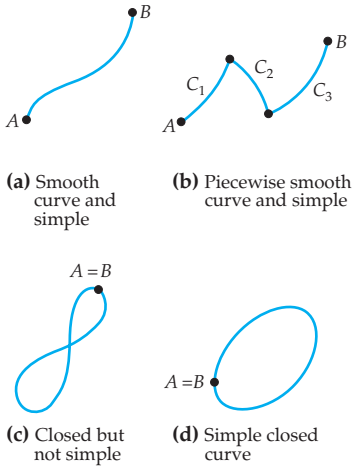
(*i*) C is a **smooth curve** if x' and y' are continuous on the closed interval $[a, b]$ and not simultaneously zero on the open interval (a, b).

(*ii*) C is a **piecewise smooth curve** if it consists of a finite number of smooth curves $C_1, C_2, \ldots, C_n$ joined end to end, that is, the terminal point of one curve C_k coinciding with the initial point of the next curve C_{k+1}.

(*iii*) C is a **simple curve** if the curve C does not cross itself except possibly at $t = a$ and $t = b$.

(*iv*) C is a **closed curve** if $A = B$.

(*v*) C is a **simple closed curve** if the curve C does not cross itself and $A = B$; that is, C is simple and closed.



(a) Smooth curve and simple

(b) Piecewise smooth curve and simple

(c) Closed but not simple

(d) Simple closed curve

FIGURE 5.1.2 Types of curves in the plane

FIGURE 5.1.2 illustrates each type of curve defined in (*i*)–(*v*).

Line Integrals in the Plane The following five steps lead to the definition of three **line integrals*** in the plane and are analogous to the five steps given prior to the definition of the definite integral.

Steps Leading to the Definition of Line Integrals

1. *Let G be a function of two real variables x and y defined at all points on a smooth curve C that lies in some region of the xy-plane. Let C be defined by the parametrization $x = x(t)$, $y = y(t)$, $a \le t \le b$.*
2. *Let P be a partition of the parameter interval $[a, b]$ into n subintervals $[t_{k-1}, t_k]$ of length $\Delta t_k = t_k - t_{k-1}$:*

$$a = t_0 < t_1 < t_2 < \cdots < t_{n-1} < t_n = b.$$

*An unfortunate choice of names. **Curve integrals** would be more appropriate.

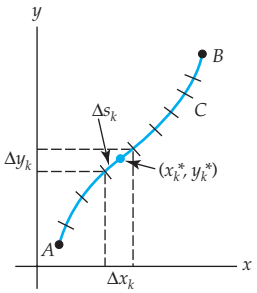


FIGURE 5.1.3 Partition of curve C into n subarcs induced by a partition P of the parameter interval $[a, b]$

The partition P induces a partition of the curve C into n subarcs of length Δs_k. Let the projection of each subarc onto the x- and y-axes have lengths Δx_k and Δy_k, respectively. See FIGURE 5.1.3.

3. *Let $\|P\|$ be the norm of the partition P of $[a, b]$, that is, the length of the longest subinterval.*
4. *Choose a point (x_k^*, y_k^*) on each subarc of C. See Figure 5.1.3.*
5. *Form n products $G(x_k^*, y_k^*)\Delta x_k$, $G(x_k^*, y_k^*)\Delta y_k$, $G(x_k^*, y_k^*)\Delta s_k$, $k = 1, 2, \ldots, n$, and then sum these products*

$$\sum_{k=1}^{n} G(x_k^*, y_k^*)\Delta x_k, \quad \sum_{k=1}^{n} G(x_k^*, y_k^*)\Delta y_k, \quad \text{and} \quad \sum_{k=1}^{n} G(x_k^*, y_k^*)\Delta s_k.$$

DEFINITION 5.1.2 Line Integrals in the Plane

(*i*) The **line integral of G along C with respect to x** is

$$\int_C G(x, y)\,dx = \lim_{\|P\|\to 0} \sum_{k=1}^{n} G(x_k^*, y_k^*)\,\Delta x_k. \tag{3}$$

(*ii*) The **line integral of G along C with respect to y** is

$$\int_C G(x, y)\,dy = \lim_{\|P\|\to 0} \sum_{k=1}^{n} G(x_k^*, y_k^*)\,\Delta y_k. \tag{4}$$

(*iii*) The **line integral of G along C with respect to arc length s** is

$$\int_C G(x, y)\,ds = \lim_{\|P\|\to 0} \sum_{k=1}^{n} G(x_k^*, y_k^*)\,\Delta s_k. \tag{5}$$

It can be proved that if G is continuous on C, then the three types of line integrals defined in (3), (4), and (5) exist. We shall assume continuity of G as matter of course. The curve C is referred to as the **path** of integration.

Method of Evaluation–C Defined Parametrically The line integrals in Definition 5.1.2 can be evaluated in two ways, depending on whether the curve C is defined by a pair of parametric equations or by an explicit function. Either way, *the basic idea is to convert a line integral to a definite integral* in a single variable. If C is smooth curve parametrized by $x = x(t)$, $y = y(t)$, $a \le t \le b$, then replace x and y in the integral by the functions $x(t)$ and $y(t)$, and the appropriate differential dx, dy, or ds by

$$dx = x'(t)\,dt, \quad dy = y'(t)\,dt, \quad \text{or} \quad ds = \sqrt{[x'(t)]^2 + [y'(t)]^2}\,dt.$$

The term $ds = \sqrt{[x'(t)]^2 + [y'(t)]^2}\,dt$ is called the differential of the arc length. In this manner each of the line integrals in Definition 5.1.2 becomes a definite integral

in which the variable of integration is the parameter t. That is,

$$\int_C G(x, y)\,dx = \int_a^b G(x(t), y(t))\,x'(t)\,dt, \tag{6}$$

$$\int_C G(x, y)\,dy = \int_a^b G(x(t), y(t))\,y'(t)\,dt, \tag{7}$$

$$\int_C G(x, y)\,ds = \int_a^b G(x(t), y(t))\sqrt{[x'(t)]^2 + [y'(t)]^2}\,dt. \tag{8}$$

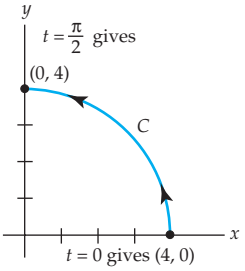


FIGURE 5.1.4 Path C of integration

EXAMPLE 1 *C* Defined Parametrically

Evaluate **(a)** $\int_C xy^2dx$, **(b)** $\int_C xy^2dy$, and **(c)** $\int_C xy^2ds$, where the path of integration C is the quarter circle defined by $x = 4\cos t$, $y = 4\sin t$, $0 \le t \le \pi/2$.

Solution The path C of integration is shown in color in **FIGURE 5.1.4**. In each of the three given line integrals, x is replaced by $4\cos t$ and y is replaced by $4\sin t$.

(a) Since $dx = -4\sin t\,dt$, we have from (6):

$$\int_C xy^2dx = \int_0^{\pi/2} (4\cos t)(4\sin t)^2(-4\sin t\,dt)$$
$$= -256\int_0^{\pi/2} \sin^3 t\cos t\,dt = -256\left[\frac{1}{4}\sin^4 t\right]_0^{\pi/2} = -64.$$

(b) Since $dy = 4\cos t\,dt$, we have from (7):

$$\int_C xy^2dy = \int_0^{\pi/2} (4\cos t)(4\sin t)^2(4\cos t\,dt)$$
$$= 256\int_0^{\pi/2} \sin^2 t\cos^2 t\,dt$$
$$= 256\int_0^{\pi/2} \frac{1}{4}\sin^2 2t\,dt$$
$$= 64\int_0^{\pi/2} \frac{1}{2}(1 - \cos 4t)\,dt = 32\left[t - \frac{1}{4}\sin 4t\right]_0^{\pi/2} = 16\pi.$$

Note in this integration we have used two trigonometric identities: $\sin 2\theta = 2\sin\theta\cos\theta$ and $\sin^2\theta = \frac{1}{2}(1 - \cos 2\theta)$.

(c) Since $ds = \sqrt{16\left(\sin^2 t + \cos^2 t\right)}\,dt = 4\,dt$, it follows from (8):

$$\int_C xy^2ds = \int_0^{\pi/2} (4\cos t)(4\sin t)^2(4\,dt)$$
$$= 256\int_0^{\pi/2} \sin^2 t\cos t\,dt = 256\left[\frac{1}{3}\sin^3 t\right]_0^{\pi/2} = \frac{256}{3}.$$

☰

Method of Evaluation–*C* Defined by a Function If the path of integration C is the graph of an explicit function $y = f(x)$, $a \le x \le b$, then we can use x as a parameter. In this situation, the differential of y is $dy = f'(x)\,dx$, and the differential of arc length is $ds = \sqrt{1 + [f'(x)]^2}\,dx$. After substituting, the three

line integrals of Definition 5.1.2 become, in turn, the definite integrals:

$$\int_C G(x, y)\,dx = \int_a^b G(x, f(x))\,dx, \tag{9}$$

$$\int_C G(x, y)\,dy = \int_a^b G(x, f(x))\,f'(x)\,dx, \tag{10}$$

$$\int_C G(x, y)\,ds = \int_a^b G(x, f(x))\sqrt{1 + [f'(x)]^2}\,dx. \tag{11}$$

A line integral along a *piecewise smooth* curve C is defined as the sum of the integrals over the various smooth curves whose union comprises C. For example, to evaluate $\int_C G(x, y)\,ds$ when C is composed of two smooth curves C_1 and C_2, we begin by writing

$$\int_C G(x, y)\,ds = \int_{C_1} G(x, y)\,ds + \int_{C_2} G(x, y)\,ds. \tag{12}$$

The integrals $\int_{C_1} G(x, y)\,ds$ and $\int_{C_2} G(x, y)\,ds$ are then evaluated in the manner given in (8) or (11).

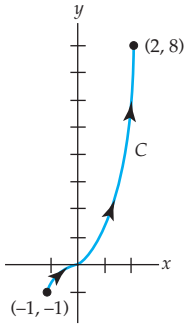


FIGURE 5.1.5 Graph of $y = x^3$ on the interval $-1 \le x \le 2$

EXAMPLE 2 *C* Defined by an Explicit Function

Evaluate $\int_C xy\,dx + x^2dy$, where C is the graph of $y = x^3$, $-1 \le x \le 2$.

Solution The curve C is illustrated in FIGURE 5.1.5 and is defined by the explicit function $y = x^3$. Hence we can use x as the parameter. Using the differential $dy = 3x^2dx$, we apply (9) and (10):

$$\begin{aligned}\int_C xy\,dx + x^2dy &= \int_{-1}^{2} x\left(x^3\right)dx + x^2\left(3x^2dx\right)\\ &= \int_{-1}^{2} 4x^4dx = \frac{4}{5}x^5\Big|_{-1}^{2} = \frac{132}{5}.\end{aligned}$$

Notation In many applications, line integrals appear as a sum $\int_C P(x, y)\,dx + \int_C Q(x, y)\,dy$. It is common practice to write this sum as one integral without parentheses as

$$\int_C P(x, y)\,dx + Q(x, y)\,dy \qquad \text{or simply} \qquad \int_C P\,dx + Q\,dy. \tag{13}$$

A line integral along a closed curve C is usually denoted by

$$\oint_C P\,dx + Q\,dy.$$

EXAMPLE 3 *C* is a Closed Curve

Evaluate $\oint_C x\,dx$, where C is the circle defined by $x = \cos t$, $y = \sin t$, $0 \le t \le 2\pi$.

Solution The differential of $x = \cos t$ is $dx = -\sin t\,dt$, and so from (6),

$$\oint_C x\,dx = \int_0^{2\pi} \cos t\,(-\sin t\,dt) = \frac{1}{2}\cos^2 t\Big|_0^{2\pi} = \frac{1}{2}[1 - 1] = 0.$$

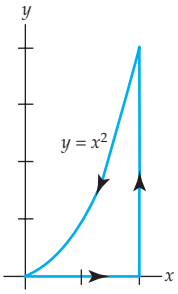


FIGURE 5.1.6 Piecewise smooth path of integration

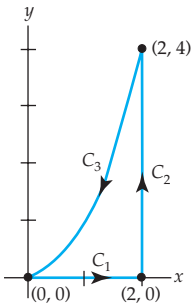


FIGURE 5.1.7 C consists of the union of C_1, C_2, and C_3

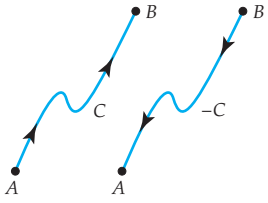


FIGURE 5.1.8 Curve C and its opposite $-C$

EXAMPLE 4 *C* is a Closed Curve

Evaluate $\oint_C y^2\,dx - x^2dy$, where C is the closed curve shown in FIGURE 5.1.6.

Solution Since C is piecewise smooth, we proceed as illustrated in (12); namely, the given integral is expressed as a sum of integrals. Symbolically, we write $\oint_C = \int_{C_1} + \int_{C_2} + \int_{C_3}$, where the C_1, C_2, and C_3 are the curves labeled in FIGURE 5.1.7. On C_1, we use x as a parameter. Since $y = 0$, $dy = 0$; therefore

$$\int_{C_1} y^2\,dx - x^2dy = \int_0^2 0\,dx - x^2(0) = 0.$$

On C_2, we use y as a parameter. From $x = 2$, $dx = 0$, we have

$$\int_{C_2} y^2\,dx - x^2dy = \int_0^4 y^2(0) - 4\,dy = -\int_0^4 4\,dy = -16.$$

On C_3, we again use x as a parameter. From $y = x^2$, we get $dy = 2x\,dx$ and so

$$\begin{aligned}\int_{C_3} y^2\,dx - x^2dy &= \int_2^0 \left(x^2\right)^2 dx - x^2\,(2x\,dx)\\ &= \int_2^0 \left(x^4 - 2x^3\right)dx = \left(\frac{1}{5}x^5 - \frac{1}{2}x^4\right)\Big|_2^0 = \frac{8}{5}.\end{aligned}$$

Hence, $$\oint_C y^2dx - x^2dy = \int_{C_1} + \int_{C_2} + \int_{C_3} = 0 + (-16) + \frac{8}{5} = -\frac{72}{5}.$$ ☰

Orientation of a Curve

In definite integration we normally assume that the interval of integration is $a \le x \le b$ and the symbol $\int_a^b f(x)\,dx$ indicates that we are integrating in the positive direction on the x-axis. Integration in the opposite direction, from $x = b$ to $x = a$, results in the negative of the original integral:

$$\int_b^a f(x)\,dx = -\int_a^b f(x)\,dx. \tag{14}$$

Line integrals possess a property similar to (14), but first we have to introduce the notion of **orientation** of the path C. If C is not a closed curve, then we say the **positive direction** on C, or that C has **positive orientation**, if we traverse C from its initial point A to its terminal point B. In other words, if $x = x(t)$, $y = y(t)$, $a \le t \le b$ are parametric equations for C, then the positive direction on C corresponds to increasing values of the parameter t. If C is traversed in the sense opposite to that of the positive orientation, then C is said to have **negative orientation**. If C has an orientation (positive or negative), then the **opposite curve**, the curve with the opposite orientation, will be denoted by the symbol $-C$. In FIGURE 5.1.8 if we assume that A and B are the initial and terminal points of the curve C, respectively, then the arrows on curve C indicate that we are traversing the curve from its initial point to its terminal point, and so C has positive orientation. The curve to the right of C that is labeled $-C$ then has negative orientation. Finally, if $-C$ denotes the curve having the opposite orientation of C, then the analogue of (14) for line integrals is

$$\int_{-C} P\,dx + Q\,dy = -\int_C P\,dx + Q\,dy, \tag{15}$$

or, equivalently

$$\int_{-C} P\,dx + Q\,dy + \int_C P\,dx + Q\,dy = 0. \tag{16}$$

For example, in part (a) of Example 1 we saw that $\int_C xy^2\,dx = -64$; we conclude from (15) that $\int_{-C} xy^2\,dx = 64$.

Note ▶ It is important to be aware that a line integral is independent of the parametrization of the curve C, provided C is given the same orientation by all sets of parametric equations defining the curve. See Problem 33 in Exercises 5.1.

EXERCISES 5.1

Answers to selected odd-numbered problems begin on page ANS-14.

In Problems 1–10, evaluate the definite integral. If necessary, review the techniques of integration in your calculus text.

1. $\displaystyle\int_{-1}^{3} x(x-1)(x+2)\,dx$

2. $\displaystyle\int_{-1}^{0} t^2\,dt + \int_{0}^{2} x^2\,dx + \int_{2}^{3} u^2\,du$

3. $\displaystyle\int_{1/2}^{1} \sin 2\pi x\,dx$

4. $\displaystyle\int_{0}^{\pi/8} \sec^2 2x\,dx$

5. $\displaystyle\int_{0}^{4} \frac{dx}{2x+1}$

6. $\displaystyle\int_{\ln 2}^{\ln 3} e^{-x}\,dx$

7. $\displaystyle\int_{2}^{4} xe^{-x/2}\,dx$

8. $\displaystyle\int_{1}^{e} \ln x\,dx$

9. $\displaystyle\int_{2}^{4} \frac{dx}{x^2-6x+5}$

10. $\displaystyle\int_{2}^{4} \frac{2x-1}{(x+3)^2}\,dx$

In Problems 11–14, evaluate the line integrals $\int_C G(x, y)\,dx$, $\int_C G(x, y)\,dy$, and $\int_C G(x, y)\,ds$ on the indicated curve C.

11. $G(x, y) = 2xy$; $x = 5\cos t$, $y = 5\sin t$, $0 \le t \le \pi/4$

12. $G(x, y) = x^3 + 2xy^2 + 2x$; $x = 2t$, $y = t^2$, $0 \le t \le 1$

13. $G(x, y) = 3x^2 + 6y^2$; $y = 2x + 1$, $-1 \le x \le 0$

14. $G(x, y) = x^2/y^3$; $2y = 3x^{3/2}$, $1 \le t \le 8$

In Problems 15–18, evaluate $\int_C (2x + y)\,dx + xy\,dy$ on the given curve from $(-1, 2)$ to $(2, 5)$.

15. $y = x + 3$

16. $y = x^2 + 1$

17.

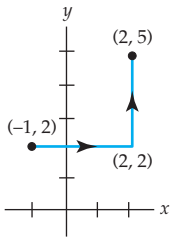


FIGURE 5.1.9 Figure for Problem 17

18.

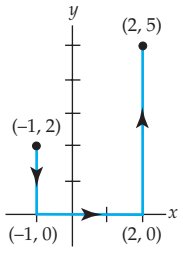


FIGURE 5.1.10 Figure for Problem 18

In Problems 19–22, evaluate $\displaystyle\int_C y\,dx + x\,dy$ on the given curve from $(0, 0)$ to $(1, 1)$.

19. $y = x^2$

20. $y = x$

21. C consists of the line segments from $(0, 0)$ to $(0, 1)$ and from $(0, 1)$ to $(1, 1)$.

22. C consists of the line segments from $(0, 0)$ to $(1, 0)$ and from $(1, 0)$ to $(1, 1)$.

23. Evaluate $\displaystyle\int_C \left(6x^2 + 2y^2\right)\,dx + 4xy\,dy$, where C is given by $x = \sqrt{t}$, $y = t$, $4 \le t \le 9$.

24. Evaluate $\int_C -y^2\,dx + xy\,dy$, where C is given by $x = 2t,\ y = t^3,\ 0 \le t \le 2$.

25. Evaluate $\int_C 2x^3y\,dx + (3x + y)\,dy$, where C is given by $x = y^2$ from $(1,\ -1)$ to $(1,\ 1)$.

26. Evaluate $\int_C 4x\,dx + 2y\,dy$, where C is given by $x = y^3 + 1$ from $(0,\ -1)$ to $(9,\ 2)$.

In Problems 27 and 28, evaluate $\oint_C (x^2 + y^2)\,dx - 2xy\,dy$ on the given closed curve.

27.

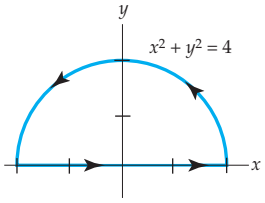


FIGURE 5.1.11 Figure for Problem 27

28.

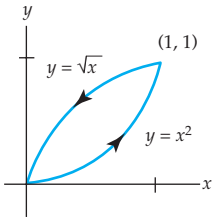


FIGURE 5.1.12 Figure for Problem 28

In Problems 29 and 30, evaluate $\oint_C x^2y^3\,dx - xy^2\,dy$ on the given closed curve.

29.

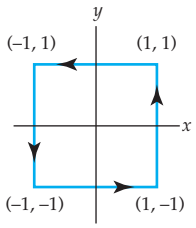


FIGURE 5.1.13 Figure for Problem 29

30.

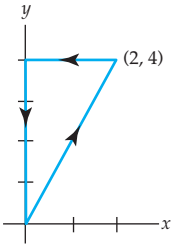


FIGURE 5.1.14 Figure for Problem 30

31. Evaluate $\oint_C (x^2 - y^2)\,ds$, where C is given by $x = 5\cos t,\ y = 5\sin t, 0 \le t \le 2\pi$.

32. Evaluate $\int_{-C} y\,dx - x\,dy$, where C is given by $x = 2\cos t,\ y = 3\sin t,\ 0 \le t \le \pi$.

33. Verify that the line integral $\int_C y^2\,dx + xy\,dy$ has the same value on C for each of the following parametrizations:

$$C : x = 2t + 1,\ y = 4t + 2,\ 0 \le t \le 1$$
$$C : x = t^2,\ y = 2t^2,\ 1 \le t \le \sqrt{3}$$
$$C : x = \ln t,\ y = 2\ln t,\ e \le t \le e^3.$$

34. Consider the three curves between $(0, 0)$ and $(2, 4)$:

$$C : x = t,\ y = 2t,\ 0 \le t \le 2$$
$$C : x = t,\ y = t^2,\ 0 \le t \le 2$$
$$C : x = 2t - 4,\ y = 4t - 8,\ 2 \le t \le 3.$$

Show that $\int_{C_1} xy\,ds = \int_{C_3} xy\,ds$, but $\int_{C_1} xy\,ds \ne \int_{C_2} xy\,ds$. Explain.

35. If $\rho(x,\ y)$ is the density of a wire (mass per unit length), then the mass of the wire is $m = \int_C \rho(x,\ y)\,ds$. Find the mass of a wire having the shape of a semicircle $x = 1 + \cos t,\ y = \sin t,\ 0 \le t \le \pi$, if the density at a point P is directly proportional to the distance from the y-axis.

36. The coordinates of the center of mass of a wire with variable density are given by $\bar{x} = M_y/m$, $\bar{y} = M_x/m$ where

$$m = \int_C \rho(x, y)\, ds, \quad M_x = \int_C y\rho(x, y)\, ds, \quad M_y = \int_C x\rho(x, y)\, ds.$$

Find the center of mass of the wire in Problem 35.

5.2 Complex Integrals

INTRODUCTION In the preceding section we reviewed two types of real integrals. We saw that the definition of the definite integral starts with a real function $y = f(x)$ that is defined on an interval on the x-axis. Because a planar curve is the two-dimensional analogue of an interval, we then generalized the definition of $\int_a^b f(x)\, dx$ to integrals of real functions of two variables defined on a curve C in the Cartesian plane. We shall see in this section that a complex integral is defined in a manner that is quite similar to that of a line integral in the Cartesian plane.

Since curves play a big part in the definition of a complex integral, we begin with a brief review of how curves are represented in the complex plane.

Curves Revisited

Suppose the continuous real-valued functions $x = x(t)$, $y = y(t)$, $a \le t \le b$, are parametric equations of a curve C in the complex plane. If we use these equations as the real and imaginary parts in $z = x + iy$, we saw in Section 2.2 that we can describe the points z on C by means of a complex-valued function of a real variable t called a **parametrization** of C:

$$z(t) = x(t) + iy(t), \quad a \le t \le b. \tag{1}$$

For example, the parametric equations $x = \cos t$, $y = \sin t$, $0 \le t \le 2\pi$, describe a unit circle centered at the origin. A parametrization of this circle is $z(t) = \cos t + i \sin t$, or $z(t) = e^{it}$, $0 \le t \le 2\pi$. See (6)–(10) in Section 2.2.

The point $z(a) = x(a) + iy(a)$ or $A = (x(a),\ y(a))$ is called the **initial point** of C and $z(b) = x(b) + iy(b)$ or $B = (x(b),\ y(b))$ is its **terminal point**. We also saw in Section 2.6 that $z(t) = x(t) + iy(t)$ could also be interpreted as a two-dimensional vector function. Consequently, $z(a)$ and $z(b)$ can be interpreted as position vectors. As t varies from $t = a$ to $t = b$ we can envision the curve C being traced out by the moving arrowhead of $z(t)$. See FIGURE 5.2.1.

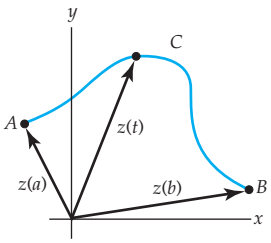


FIGURE 5.2.1 $z(t) = x(t) + iy(t)$ as a position vector

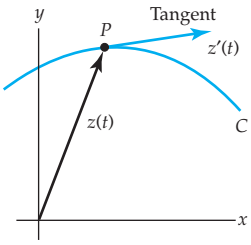


FIGURE 5.2.2 $z'(t) = x'(t) + iy'(t)$ as a tangent vector

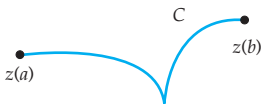
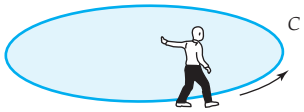


FIGURE 5.2.3 Curve C is not smooth since it has a cusp.

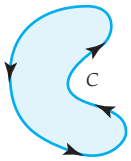
Contours

The notions of curves in the complex plane that are smooth, piecewise smooth, simple, closed, and simple closed are easily formulated in terms of the vector function (1). Suppose the derivative of (1) is $z'(t) = x'(t) + iy'(t)$. We say a curve C in the complex plane is **smooth** if $z'(t)$ is continuous and never zero in the interval $a \le t \le b$. As shown FIGURE 5.2.2, since the vector $z'(t)$ is not zero at any point P on C, the vector $z'(t)$ is tangent to C at P. In other words, a smooth curve has a continuously turning tangent; put yet another way, a smooth curve can have no sharp corners or cusps. See FIGURE 5.2.3. A **piecewise smooth curve** C has a continuously turning tangent, except possibly at the points where the component smooth curves $C_1, C_2, \ldots, C_n$ are joined together. A curve C in the complex plane is said to be a **simple** if $z(t_1) \ne z(t_2)$ for $t_1 \ne t_2$, except possibly for $t = a$ and $t = b$. C is a **closed curve** if $z(a) = z(b)$. C is a **simple closed curve** if $z(t_1) \ne z(t_2)$ for $t_1 \ne t_2$ and $z(a) = z(b)$. In complex analysis, a piecewise smooth curve C is called a **contour** or **path**.

Just as we did in the preceding section, we define the **positive direction** on a contour C to be the direction on the curve corresponding to increasing values of



(a) Positive direction



(b) Positive direction

FIGURE 5.2.4 Interior of each curve is to the left.

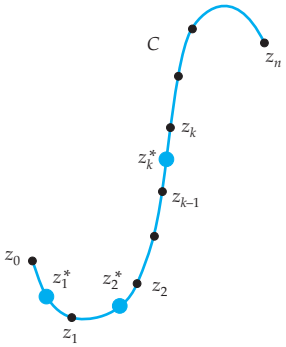


FIGURE 5.2.5 Partition of curve C into n subarcs is induced by a partition P of the parameter interval $[a, b]$.

the parameter t. It is also said that the curve C has **positive orientation**. In the case of a simple closed curve C, the positive direction roughly corresponds to the counterclockwise direction or the direction that a person must walk on C in order to keep the interior of C to the left. For example, the circle $z(t) = e^{it}$, $0 \le t \le 2\pi$, has positive orientation. See FIGURE 5.2.4. The **negative direction** on a contour C is the direction opposite the positive direction. If C has an orientation, the **opposite curve**, that is, a curve with opposite orientation, is denoted by $-C$. On a simple closed curve, the negative direction corresponds to the clockwise direction.

Complex Integral

An integral of a function f of a complex variable z that is defined on a contour C is denoted by $\int_C f(z)\,dz$ and is called a **complex integral**. The following list of assumptions is a prelude to the definition of a complex integral. For the sake of comparison, you are encouraged to review the lists given prior to Definitions 5.1.1 and 5.1.2. Also, look over FIGURE 5.2.5 as you read this new list.

Steps Leading to the Definition of the Complex Integral

1. *Let f be a function of a complex variable z defined at all points on a smooth curve C that lies in some region of the complex plane. Let C be defined by the parametrization $z(t) = x(t) + iy(t)$, $a \le t \le b$.*
2. *Let P be a partition of the parameter interval $[a, b]$ into n subintervals $[t_{k-1}, t_k]$ of length $\Delta t_k = t_k - t_{k-1}$:*

$$a = t_0 < t_1 < t_2 < \cdots < t_{n-1} < t_n = b.$$

The partition P induces a partition of the curve C into n subarcs whose initial and terminal points are the pairs of numbers

$$\begin{aligned} z_0 &= x(t_0) + iy(t_0), & z_1 &= x(t_1) + iy(t_1),\\ z_1 &= x(t_1) + iy(t_1), & z_2 &= x(t_2) + iy(t_2),\\ &\vdots & &\vdots\\ z_{n-1} &= x(t_{n-1}) + iy(t_{n-1}), & z_n &= x(t_n) + iy(t_n). \end{aligned}$$

Let $\Delta z_k = z_k - z_{k-1}$, $k = 1, 2, \ldots, n$. See Figure 5.2.5.

3. *Let $\|P\|$ be the norm of the partition P of $[a, b]$, that is, the length of the longest subinterval.*
4. *Choose a point $z_k^* = x_k^* + iy_k^*$ on each subarc of C. See Figure 5.2.5.*
5. *Form n products $f(z_k^*)\Delta z_k$, $k = 1, 2, \ldots, n$, and then sum these products:*

$$\sum_{k=1}^{n} f(z_k^*)\Delta z_k.$$

DEFINITION 5.2.1 Complex Integral

The **complex integral** of f on C is

$$\int_C f(z)\,dz = \lim_{\|P\|\to 0} \sum_{k=1}^{n} f(z_k^*)\,\Delta z_k. \tag{2}$$

General assumptions throughout this text ▶

If the limit in (2) exists, then f is said to be **integrable** on C. The limit exists whenever f is continuous at all points on C and C is either smooth or piecewise smooth. Consequently we shall, hereafter, assume these conditions as a matter of course. Moreover, we will use the notation $\oint_C f(z)\,dz$ to represent a complex integral around a *positively oriented* closed curve C. When it is important to distinguish the direction of integration around a closed curve, we will employ the notations

$$\oint_C f(z)\,dz \qquad \text{and} \qquad \oint_C f(z)\,dz$$

to denote integration in the positive and negative directions, respectively.

Note ▶

The point having been made that the definitions of real integrals discussed in Section 5.1 and Definition 5.2.1 are formally the same, we shall from now on refer to a complex integral $\int_C f(z)\,dz$ by its more common name, **contour integral**.

Complex-Valued Function of a Real Variable Before turning to the properties of contour integrals and the all-important question of how to evaluate a contour integral, we need to digress briefly to enlarge upon the concept of a **complex-valued function of a real variable** introduced in the *Remarks* in Section 2.1. As already mentioned, a parametrization of a curve C of the form given in (1) is a case in point. Let's consider another simple example. If t represents a real variable, then the output of the function $f(t) = (2t + i)^2$ is a complex number. For $t = 2$,

$$f(2) = (4 + i)^2 = 16 + 8i + i^2 = 15 + 8i.$$

In general, if f_1 and f_2 are real-valued functions of a real variable t (that is, real functions), then $f(t) = f_1(t) + if_2(t)$ is a complex-valued function of a real variable t. What we are really interested in at the moment is the definite integral $\int_a^b f(t)\,dt$, in other words, the integration of complex-valued function $f(t) = f_1(t) + if_2(t)$ of real variable t carried out over a real interval. Continuing with the specific function $f(t) = (2t + i)^2$ it seems logical to write on, say, the interval $0 \le t \le 1$,

$$\int_0^1 (2t + i)^2 dt = \int_0^1 \left(4t^2 - 1 + 4ti\right) dt = \int_0^1 (4t^2 - 1)dt + i\int_0^1 4t\,dt. \qquad (3)$$

The integrals $\int_0^1 (4t^2 - 1)dt$ and $\int_0^1 4t\,dt$ in (3) are real, and so one would be inclined to call them the real and imaginary parts of $\int_0^1 (2t + i)^2 dt$. Each of these real integrals can be evaluated using the fundamental theorem of calculus ((1) of Section 5.1):

$$\int_0^1 (4t^2 - 1)dt = \left(\frac{4}{3}t^3 - t\right)\Big|_0^1 = \frac{1}{3} \qquad \text{and} \qquad \int_0^1 4t\,dt = 2t^2\Big|_0^1 = 2.$$

Thus (3) becomes $\int_0^1 (2t + i)^2 dt = \frac{1}{3} + 2i$.

Since the preceding integration seems very ordinary and routine, we give the following generalization. If f_1 and f_2 are real-valued functions of a real variable t continuous on a common interval $a \le t \le b$, then we *define* the integral of the complex-valued function $f(t) = f_1(t) + if_2(t)$ on $a \le t \le b$ in terms of the definite integrals of the real and imaginary parts of f:

A review of basic techniques of integration that might be used to evaluate these real integrals can be found in Section 5.1 of the student study guide. ▶

$$\int_a^b f(t)\,dt = \int_a^b f_1(t)\,dt + i\int_a^b f_2(t)\,dt. \qquad (4)$$

The continuity of f_1 and f_2 on $[a, b]$ guarantees that both $\int_a^b f_1(t)\,dt$ and $\int_a^b f_2(t)\,dt$ exist.

All of the following familiar properties of integrals can be proved directly from the definition given in (4). If $f(t) = f_1(t) + if_2(t)$ and $g(t) = g_1(t) + ig_2(t)$ are complex-valued functions of a real variable t continuous on an interval $a \le t \le b$, then

$$\int_a^b k\, f(t)\, dt = k \int_a^b f(t)\, dt, \quad k \text{ a complex constant}, \tag{5}$$

These properties are important in the evaluation of contour integrals. They will often be used without mention. ▶

$$\int_a^b (f(t) + g(t))\, dt = \int_a^b f(t)\, dt + \int_a^b g(t)\, dt, \tag{6}$$

$$\int_a^b f(t)\, dt = \int_a^c f(t)\, dt + \int_c^b f(t)\, dt, \tag{7}$$

$$\int_b^a f(t)\, dt = -\int_a^b f(t)\, dt. \tag{8}$$

In (7) we choose to assume that the real number c is in the interval $[a, b]$.

We now resume our discussion of contour integrals.

Evaluation of Contour Integrals To facilitate the discussion on how to evaluate a contour integral $\int_C f(z)\, dz$, let us write (2) in an abbreviated form. If we use $u + iv$ for $f(z)$, $\Delta x + i\Delta y$ for Δz, lim for $\lim_{\|P\| \to 0}$, $\sum$ for $\sum_{k=1}^{n}$ and then suppress all subscripts, (2) becomes

$$\begin{aligned} \int_C f(z)dz &= \lim \sum (u + iv)(\Delta x + i\Delta y) \\ &= \lim \left[\sum (u\Delta x - v\Delta y) + i \sum (v\Delta x + u\Delta y) \right]. \end{aligned}$$

The interpretation of the last line is

$$\int_C f(z)\, dz = \int_C u\, dx - v\, dy + i \int_C v\, dx + u\, dy. \tag{9}$$

See Definition 5.1.2. In other words, the real and imaginary parts of a contour integral $\int_C f(z)\, dz$ are a pair of real line integrals $\int_C u\, dx - v\, dy$ and $\int_C v\, dx + u\, dy$. Now if $x = x(t)$, $y = y(t)$, $a \le t \le b$ are parametric equations of C, then $dx = x'(t)\, dt$, $dy = y'(t)\, dt$. By replacing the symbols x, y, dx, and dy by $x(t)$, $y(t)$, $x'(t)\, dt$, and $y'(t)\, dt$, respectively, the right side of (9) becomes

$$\overbrace{\int_a^b \left[u(x(t), y(t))\, x'(t) - v(x(t), y(t))\, y'(t) \right] dt}^{\int_C u dx - v dy} + i \overbrace{\int_a^b \left[v(x(t), y(t))\, x'(t) + u(x(t), y(t))\, y'(t) \right] dt}^{\int_C v dx + u dy}. \tag{10}$$

If we use the complex-valued function (1) to describe the contour C, then (10) is the same as $\int_a^b f(z(t))\, z'(t)\, dt$ when the integrand

$$f(z(t))\, z'(t) = [u(x(t), y(t)) + iv(x(t), y(t))] \left[x'(t) + iy'(t) \right]$$

is multiplied out and $\int_a^b f(z(t))\, z'(t)\, dt$ is expressed in terms of its real and imaginary parts. Thus we arrive at a practical means of evaluating a contour integral.

THEOREM 5.2.1 Evaluation of a Contour Integral

If f is continuous on a smooth curve C given by the parametrization $z(t) = x(t) + iy(t)$, $a \le t \le b$, then

$$\int_C f(z)\,dz = \int_a^b f(z(t))\,z'(t)\,dt. \tag{11}$$

The foregoing results in (10) and (11) bear repeating—this time in somewhat different words. Suppose $z(t) = x(t) + iy(t)$ and $z'(t) = x'(t) + iy'(t)$. Then the integrand $f(z(t))z'(t)$ is a complex-valued function of a real variable t. Hence the integral $\int_a^b f(z(t))z'(t)\,dt$ is evaluated in the manner defined in (4). The next example illustrates the method.

EXAMPLE 1 Evaluating a Contour Integral

Evaluate $\int_C \bar{z}\,dz$, where C is parametrized by $z(t) = 3t + it^2$, $-1 \le t \le 4$.

Solution With the identification $f(z) = \bar{z}$ we have $f(z(t)) = \overline{3t + it^2} = 3t - it^2$. Also, $z'(t) = 3 + 2it$, and so by (11) the integral is

$$\int_C \bar{z}\,dz = \int_{-1}^{4} (3t - it^2)(3 + 2it)\,dt = \int_{-1}^{4} \left[2t^3 + 9t + 3t^2 i\right] dt.$$

Now in view of (4), the last integral is the same as

$$\begin{aligned}\int_C \bar{z}\,dz &= \int_{-1}^{4} (2t^3 + 9t)\,dt + i\int_{-1}^{4} 3t^2\,dt \\ &= \left(\frac{1}{2}t^4 + \frac{9}{2}t^2\right)\Big|_{-1}^{4} + it^3\Big|_{-1}^{4} = 195 + 65i.\end{aligned}$$

EXAMPLE 2 Evaluating a Contour Integral

Evaluate $\oint_C \frac{1}{z}\,dz$, where C is the circle $x = \cos t$, $y = \sin t$, $0 \le t \le 2\pi$.

Solution In this case $z(t) = \cos t + i \sin t = e^{it}$, $z'(t) = ie^{it}$, and $f(z(t)) = \dfrac{1}{z(t)} = e^{-it}$. Hence,

$$\oint_C \frac{1}{z}\,dz = \int_0^{2\pi} (e^{-it})\,ie^{it}\,dt = i\int_0^{2\pi} dt = 2\pi i.$$

As discussed in Section 5.1, for some curves the real variable x itself can be used as the parameter. For example, to evaluate $\int_C (8x^2 - iy)\,dz$ on the line segment $y = 5x$, $0 \le x \le 2$, we write $z = x + 5xi$, $dz = (1 + 5i)dx$, $\int_C (8x^2 - iy)\,dz = \int_0^2 (8x^2 - 5ix)(1 + 5i)\,dx$, and then integrate in the usual manner:

$$\begin{aligned}\int_C (8x^2 - iy)\,dz &= (1 + 5i)\int_0^2 (8x^2 - 5ix)\,dx \\ &= (1 + 5i)\left[\frac{8}{3}x^3\right]_0^2 - (1 + 5i)i\left[\frac{5}{2}x^2\right]_0^2 = \frac{214}{3} + \frac{290}{3}i.\end{aligned}$$

In general, if x and y are related by means of a continuous real function $y = f(x)$, then the corresponding curve C in the complex plane can be parametrized by $z(x) = x + if(x)$. Equivalently, we can let $x = t$ so that a set of parametric equations for C is $x = t$, $y = f(t)$.

Properties The following properties of contour integrals are analogous to the properties of real line integrals as well as the properties listed in (5)–(8).

THEOREM 5.2.2 Properties of Contour Integrals

Suppose the functions f and g are continuous in a domain D, and C is a smooth curve lying entirely in D. Then

(*i*) $\int_C kf(z)\,dz = k\int_C f(z)\,dz$, k a complex constant.

(*ii*) $\int_C [f(z) + g(z)]\,dz = \int_C f(z)\,dz + \int_C g(z)\,dz$.

(*iii*) $\int_C f(z)\,dz = \int_{C_1} f(z)\,dz + \int_{C_2} f(z)\,dz$, where C consists of the smooth curves C_1 and C_2 joined end to end.

(*iv*) $\int_{-C} f(z)\,dz = -\int_C f(z)\,dz$, where $-C$ denotes the curve having the opposite orientation of C.

The four parts of Theorem 5.2.2 also hold if C is a piecewise smooth curve in D.

EXAMPLE 3 *C* Is a Piecewise Smooth Curve

Evaluate $\int_C (x^2 + iy^2)\,dz$, where C is the contour shown in FIGURE 5.2.6.

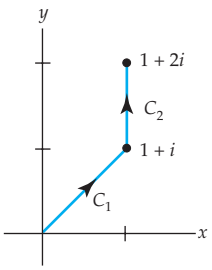


FIGURE 5.2.6 Contour C is piecewise smooth.

Solution In view of Theorem 5.2.2(*iii*) we write

$$\int_C (x^2 + iy^2)\,dz = \int_{C_1} (x^2 + iy^2)\,dz + \int_{C_2} (x^2 + iy^2)\,dz.$$

Since the curve C_1 is defined by $y = x$, $0 \le x \le 1$, it makes sense to use x as a parameter. Therefore, $z(x) = x + ix$, $z'(x) = 1 + i$, $f(z) = x^2 + iy^2$, $f(z(x)) = x^2 + ix^2$, and

$$\int_{C_1} (x^2 + iy^2)\,dz = \int_0^1 \overbrace{(x^2 + ix^2)}^{(1+i)x^2}(1 + i)\,dx$$

$$= (1 + i)^2 \int_0^1 x^2 dx = \frac{(1 + i)^2}{3} = \frac{2}{3}i. \tag{12}$$

The curve C_2 is defined by $x = 1$, $1 \le y \le 2$. If we use y as a parameter, then $z(y) = 1 + iy$, $z'(y) = i$, $f(z(y)) = 1 + iy^2$, and

$$\int_{C_2} (x^2 + iy^2)\,dz = \int_1^2 (1 + iy^2)i\,dy = -\int_1^2 y^2\,dy + i\int_1^2 dy = -\frac{7}{3} + i. \tag{13}$$

Combining (10) and (13) gives $\int_C (x^2 + iy^2)\,dz = \frac{2}{3}i + (-\frac{7}{3} + i) = -\frac{7}{3} + \frac{5}{3}i$. ☰

There are times in the application of complex integration that it is useful to find an upper bound for the modulus or absolute value of a contour integral. In the next

theorem we use the fact that the length of a plane curve is $L = \int_a^b \sqrt{[x'(t)]^2 + [y'(t)]^2}\, dt$. But if $z'(t) = x'(t) + iy'(t)$, then $|z'(t)| = \sqrt{[x'(t)]^2 + [y'(t)]^2}$ and, consequently, $L = \int_a^b |z'(t)|\, dt$.

THEOREM 5.2.3 A Bounding Theorem

If f is continuous on a smooth curve C and if $|f(z)| \le M$ for all z on C, then $\left|\int_C f(z)\, dz\right| \le ML$, where L is the length of C.

Proof It follows from the form of the triangle inequality given in (11) of Section 1.2 that

$$\left|\sum_{k=1}^{n} f(z_k^*)\Delta z_k\right| \le \sum_{k=1}^{n} |f(z_k^*)|\, |\Delta z_k| \le M\sum_{k=1}^{n} |\Delta z_k|. \tag{14}$$

Because $|\Delta z_k| = \sqrt{(\Delta x_k)^2 + (\Delta y_k)^2}$, we can interpret $|\Delta z_k|$ as the length of the chord joining the points z_k and z_{k-1} on C. Moreover, since the sum of the lengths of the chords cannot be greater than the length L of C, the inequality (14) continues as $\left|\sum_{k=1}^{n} f(z_k^*)\Delta z_k\right| \le ML$. Finally, the continuity of f guarantees that $\int_C f(z)\, dz$ exists, and so if we let $\|P\| \to 0$, the last inequality yields $\left|\int_C f(z)\, dz\right| \le ML$. ☰

Theorem 5.2.3 is used often in the theory of complex integration and is sometimes referred to as the ***ML*-inequality**. It follows from the discussion on page 114 of Section 3.1 that since f is continuous on the contour C, the bound M for the values $f(z)$ in Theorem 5.2.3 will always exist.

EXAMPLE 4 A Bound for a Contour Integral

Find an upper bound for the absolute value of $\oint_C \frac{e^z}{z+1}\, dz$ where C is the circle $|z| = 4$.

Solution First, the length L (circumference) of the circle of radius 4 is 8π. Next, from the inequality (7) of Section 1.2, it follows for all points z on the circle that $|z + 1| \ge |z| - 1 = 4 - 1 = 3$. Thus

$$\left|\frac{e^z}{z+1}\right| \le \frac{|e^z|}{|z| - 1} = \frac{|e^z|}{3}. \tag{15}$$

In addition, $|e^z| = |e^x(\cos y + i \sin y)| = e^x$. For points on the circle $|z| = 4$, the maximum that $x = \text{Re}(z)$ can be is 4, and so (15) yields

$$\left|\frac{e^z}{z+1}\right| \le \frac{e^4}{3}.$$

From the *ML*-inequality (Theorem 5.2.3) we have

$$\left|\oint_C \frac{e^z}{z+1}\, dz\right| \le \frac{8\pi e^4}{3}.$$

☰

Remarks

There is no unique parametrization for a contour C. You should verify that

$$z(t) = e^{it} = \cos t + i \sin t,\ 0 \le t \le 2\pi$$
$$z(t) = e^{2\pi it} = \cos 2\pi t + i \sin 2\pi t,\ 0 \le t \le 1$$
$$z(t) = e^{\pi it/2} = \cos \frac{\pi t}{2} + i \sin \frac{\pi t}{2},\ 0 \le t \le 4$$

are all parametrizations, oriented in the positive direction, for the unit circle $|z| = 1$.

EXERCISES 5.2

Answers to selected odd-numbered problems begin on page ANS-14.

In Problems 1–16, evaluate the given integral along the indicated contour.

1. $\int_C (z+3)\,dz$, where C is $x = 2t$, $y = 4t - 1$, $1 \le t \le 3$

2. $\int_C (2\bar{z} - z)\,dz$, where C is $x = -t$, $y = t^2 + 2$, $0 \le t \le 2$

3. $\int_C z^2\,dz$, where C is $z(t) = 3t + 2it$, $-2 \le t \le 2$

4. $\int_C (3z^2 - 2z)\,dz$, where C is $z(t) = t + it^2$, $0 \le t \le 1$

5. $\int_C \frac{z+1}{z}\,dz$, where C is the right half of the circle $|z| = 1$ from $z = -i$ to $z = i$

6. $\int_C |z|^2\,dz$, where C is $x = t^2$, $y = 1/t$, $1 \le t \le 2$

7. $\oint_C \mathrm{Re}(z)\,dz$, where C is the circle $|z| = 1$

8. $\oint_C \left(\frac{1}{(z+i)^3} - \frac{5}{z+i} + 8\right) dz$, where C is the circle $|z + i| = 1$

9. $\int_C (x^2 + iy^3)\,dz$, where C is the straight line from $z = 1$ to $z = i$

10. $\int_C (x^2 - iy^3)\,dz$, where C is the lower half of the circle $|z| = 1$ from $z = -1$ to $z = 1$

11. $\int_C e^z\,dz$, where C is the polygonal path consisting of the line segments from $z = 0$ to $z = 2$ and from $z = 2$ to $z = 1 + \pi i$

12. $\int_C \sin z\,dz$, where C is the polygonal path consisting of the line segments from $z = 0$ to $z = 1$ and from $z = 1$ to $z = 1 + i$

13. $\int_C \mathrm{Im}\,(z - i)\,dz$, where C is the polygonal path consisting of the circular arc along $|z| = 1$ from $z = 1$ to $z = i$ and the line segment from $z = i$ to $z = -1$

14. $\int_C dz$, where C is the left half of the ellipse $\frac{1}{36}x^2 + \frac{1}{4}y^2 = 1$ from $z = 2i$ to $z = -2i$

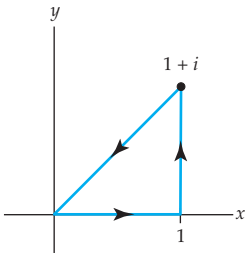


FIGURE 5.2.7 Figure for Problems 17–20

15. $\oint_C ze^z\,dz$, where C is the square with vertices $z = 0$, $z = 1$, $z = 1 + i$, and $z = i$

16. $\int_C f(z)\,dz$, where $f(z) = \begin{cases} 2, & x < 0 \\ 6x, & x > 0 \end{cases}$ and C is the parabola $y = x^2$ from $z = -1 + i$ to $z = 1 + i$

In Problems 17–20, evaluate the given integral along the contour C given in **FIGURE 5.2.7**.

17. $\oint_C x\,dz$

18. $\oint_C (2z - 1)\,dz$

19. $\oint_C z^2\,dz$

20. $\oint_C \bar{z}^2\,dz$

In Problems 21–24, evaluate $\int_C (z^2 - z + 2)\,dz$ from i to 1 along the contour C given in the figures.

21.

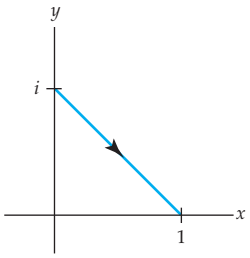


FIGURE 5.2.8 Figure for Problem 21

22.

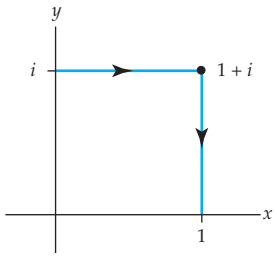


FIGURE 5.2.9 Figure for Problem 22

23.

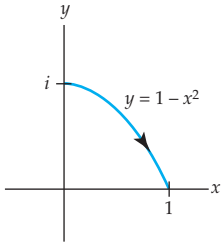


FIGURE 5.2.10 Figure for Problem 23

24.

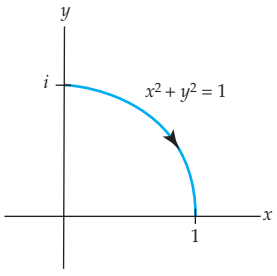


FIGURE 5.2.11 Figure for Problem 24

In Problems 25–28, find an upper bound for the absolute value of the given integral along the indicated contour.

25. $\oint_C \dfrac{e^z}{z^2 + 1}\,dz$, where C is the circle $|z| = 5$

26. $\int_C \dfrac{1}{z^2 - 2i}\,dz$, where C is the right half of the circle $|z| = 6$ from $z = -6i$ to $z = 6i$

27. $\int_C (z^2 + 4)\,dz$, where C is the line segment from $z = 0$ to $z = 1 + i$

28. $\int_C \dfrac{1}{z^3}\,dz$, where C is one-quarter of the circle $|z| = 4$ from $z = 4i$ to $z = 4$

Focus on Concepts

29. (**a**) Use Definition 5.2.1 to show for any smooth curve C between z_0 and z_n that $\int_C dz = z_n - z_0$.

(**b**) Use the result in part (a) to verify your answer to Problem 14.

(**c**) What is $\oint_C dz$ when C is a simple closed curve?

30. Use Definition 5.2.1 to show for any smooth curve C between z_0 and z_n that $\int_C z\,dz = \frac{1}{2}(z_n^2 - z_0^2)$. [*Hint*: The integral exists. So choose $z_k^* = z_k$ and $z_k^* = z_{k-1}$.]

31. Use the results of Problems 29 and 30 to evaluate $\int_C (6z + 4)\,dz$ where C is:

(**a**) The straight line from $1 + i$ to $2 + 3i$.

(**b**) The closed contour $x^4 + y^4 = 4$.

32. Find an upper bound for the absolute value of the integral $\displaystyle\int_C \frac{1}{z^2 + 1}\,dz$, where the contour C is the line segment from $z = 3$ to $z = 3 + i$. Use the fact that $|z^2 + 1| = |z - i|\,|z + i|$ where $|z - i|$ and $|z + i|$ represent, respectively, the distances from i and $-i$ to points z on C.

33. Find an upper bound for the absolute value of the integral $\displaystyle\oint_C \frac{z^2}{(z^2 + 9)^2}dz$, where C is a circle of radius $R > 3$ centered at the origin. Use the fact that $|z^2 + 9| = |z - 3i|\,|z + 3i|$ and apply the triangle inequality (10) of Section 1.2 to each factor.

34. Find an upper bound for the absolute value of the integral $\int_C \text{Ln}(z + 3)\,dz$, where the contour C is the line segment from $z = 3i$ to $z = 4 + 3i$.

Computer Lab Assignments

A CAS can be used to evaluate contour integrals by the method given in (11). For example, in *Mathematica* a complex function $f(z)$ and a parametrization $g(t)$ of a contour C can be defined as described on page 59. In order to now compute the contour integral $\int_C f(z)\,dz$ use the syntax **Integrate[f[g[t]]*g'[t],{t,a,b}]**. In Problems 35–38, use a CAS to evaluate the given contour integral

35. The integral in Problem 2.

36. The integral in Problem 4.

37. The integral in Problem 6.

38. The integral in Problem 8.

5.3 Cauchy-Goursat Theorem

INTRODUCTION In this section we shall concentrate on contour integrals, where the contour C is a simple closed curve with a positive (counterclockwise) orientation. Specifically, we shall see that when f is analytic in a special kind of domain D, the value of the contour integral $\oint_C f(z)\,dz$ is the same for *any* simple closed curve C that lies entirely within D. This theorem, called the **Cauchy-Goursat theorem**, is one of *the* fundamental results in complex analysis.

Preliminary to discussing the Cauchy-Goursat theorem and some of its ramifications, we need to distinguish two kinds of domains in the complex plain: simply connected and multiply connected.

Simply and Multiply Connected Domains Recall from Section 1.5 that a domain is an open connected set in the complex plane. We say that a domain D is **simply connected** if every simple closed contour C lying entirely in D

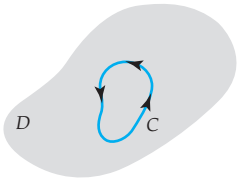


FIGURE 5.3.1 Simply connected domain D

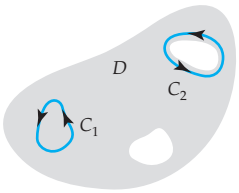


FIGURE 5.3.2 Multiply connected domain D

can be shrunk to a point without leaving D. See FIGURE 5.3.1. In other words, if we draw any simple closed contour C so that it lies entirely within a simply connected domain, then C encloses only points of the domain D. Expressed yet another way, a simply connected domain has no "holes" in it. The entire complex plane is an example of a simply connected domain; the annulus defined by $1 < |z| < 2$ is not simply connected. (Why?) A domain that is not simply connected is called a **multiply connected domain**; that is, a multiply connected domain has "holes" in it. Note in FIGURE 5.3.2 that if the curve C_2 enclosing the "hole" were shrunk to a point, the curve would have to leave D eventually. We call a domain with one "hole" **doubly connected**, a domain with two "holes" **triply connected**, and so on. The open disk defined by $|z| < 2$ is a simply connected domain; the open circular annulus defined by $1 < |z| < 2$ is a doubly connected domain.

Cauchy's Theorem

In 1825 the French mathematician Louis-Augustin Cauchy proved one the most important theorems in complex analysis.

Cauchy's Theorem

Suppose that a function f is analytic in a simply connected domain D and that f' is continuous in D. Then for every simple closed contour C in D, $\oint_C f(z)\,dz = 0$. (1)

Cauchy's Proof of (1) The proof of this theorem is an immediate consequence of Green's theorem in the plane and the Cauchy-Riemann equations. Recall from calculus that if C is a positively oriented, piecewise smooth, simple closed curve forming the boundary of a region R within D, and if the real-valued functions $P(x, y)$ and $Q(x, y)$ along with their first-order partial derivatives are continuous on a domain that contains C and R, then

Green's theorem expresses a real line integral as a double integral ▶

$$\oint_C P\,dx + Q\,dy = \iint_R \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y}\right) dA. \tag{2}$$

Now in the statement (1) we have assumed that f' is continuous throughout the domain D. As a consequence, the real and imaginary parts of $f(z) = u + iv$ and their first partial derivatives are continuous throughout D. By (9) of Section 5.2 we write $\oint_C f(z)\,dz$ in terms of real line integrals and apply Green's theorem (2) to each line integral:

$$\begin{aligned}\oint_C f(z)\,dz &= \oint_C u(x, y)\,dx - v(x, y)\,dy + i\oint_C v(x, y)\,dx + u(x, y)\,dy \\ &= \iint_R \left(-\frac{\partial v}{\partial x} - \frac{\partial u}{\partial y}\right) dA + i\iint_R \left(\frac{\partial u}{\partial x} - \frac{\partial v}{\partial y}\right) dA. \end{aligned}\tag{3}$$

Because f is analytic in D, the real functions u and v satisfy the Cauchy-Riemann equations, $\partial u/\partial x = \partial v/\partial y$ and $\partial u/\partial y = -\partial v/\partial x$, at every point in D. Using the Cauchy-Riemann equations to replace $\partial u/\partial y$ and $\partial u/\partial x$ in (3) shows that

$$\begin{aligned}\oint_C f(z)\,dz &= \iint_R \left(-\frac{\partial v}{\partial x} + \frac{\partial v}{\partial x}\right) dA + i\iint_R \left(\frac{\partial v}{\partial y} - \frac{\partial v}{\partial y}\right) dA \\ &= \iint_R (0)\,dA + i\iint_R (0)\,dA = 0.\end{aligned}$$

This completes the proof. ☰

In 1883 the French mathematician Edouard Goursat proved that the assumption of continuity of f' is not necessary to reach the conclusion of Cauchy's theorem. The resulting modified version of Cauchy's theorem is known today as the **Cauchy-Goursat theorem**. As one might expect, with fewer hypotheses, the proof of this version of Cauchy's theorem is more complicated than the one just presented. A form of the proof devised by Goursat is outlined in Appendix II.

THEOREM 5.3.1 Cauchy-Goursat Theorem

Suppose that a function f is analytic in a simply connected domain D. Then for every simple closed contour C in D, $\oint_C f(z)\,dz = 0$.

Since the interior of a simple closed contour is a simply connected domain, the Cauchy-Goursat theorem can be stated in the slightly more practical manner:

- *If f is analytic at all points within and on a simple closed contour C, then $\oint_C f(z)\,dz = 0$.* (4)

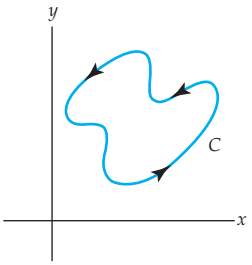


FIGURE 5.3.3 Contour for Example 1

EXAMPLE 1 Applying the Cauchy-Goursat Theorem

Evaluate $\oint_C e^z\,dz$, where the contour C is shown in FIGURE 5.3.3.

Solution The function $f(z) = e^z$ is entire and consequently is analytic at all points within and on the simple closed contour C. It follows from the form of the Cauchy-Goursat theorem given in (4) that $\oint_C e^z\,dz = 0$ ☰

The point of Example 1 is that $\oint_C e^z\,dz = 0$ for *any* simple closed contour in the complex plane. Indeed, it follows that for any simple closed contour C and any entire function f, such as $f(z) = \sin z$, $f(z) = \cos z$, and $p(z) = a_n z^n + a_{n-1}z^{n-1} + \cdots + a_1 z + a_0$, $n = 0, 1, 2, \ldots$, that

$$\oint_C \sin z\,dz = 0, \quad \oint_C \cos z\,dz = 0, \quad \oint_C p(z)\,dz = 0,$$

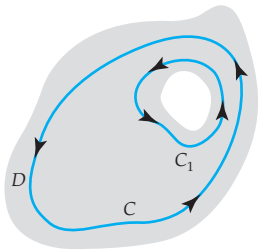
and so on.

EXAMPLE 2 Applying the Cauchy-Goursat Theorem

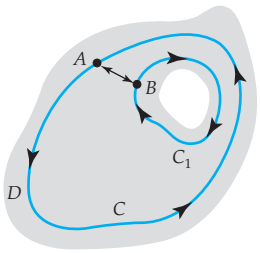
Evaluate $\oint_C \frac{dz}{z^2}$, where the contour C is the ellipse $(x-2)^2 + \frac{1}{4}(y-5)^2 = 1$.

Solution The rational function $f(z) = 1/z^2$ is analytic everywhere except at $z = 0$. But $z = 0$ is not a point interior to or on the simple closed elliptical contour C. Thus, from (4) we have that $\oint_C \frac{dz}{z^2} = 0$. ☰

Multiply Connected Domains If f is analytic in a multiply connected domain D then we cannot conclude that $\oint_C f(z)\,dz = 0$ for every simple closed contour C in D. To begin, suppose that D is a doubly connected domain and C and C_1 are simple closed contours such that C_1 surrounds the "hole" in the domain



(a)



(b)

FIGURE 5.3.4 Doubly connected domain D

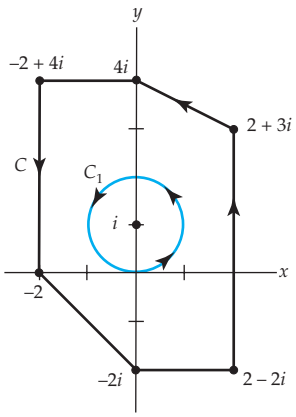


FIGURE 5.3.5 We use the simpler contour C_1 in Example 3

and is interior to C. See **FIGURE 5.3.4(a)**. Suppose, also, that f is analytic on each contour and at each point interior to C but exterior to C_1. By introducing the crosscut AB shown in Figure 5.3.4(b), the region bounded between the curves is now simply connected. From (*iv*) of Theorem 5.2.2, the integral from A to B has the opposite value of the integral from B to A, and so from (4) we have

$$\oint_C f(z)\,dz + \int_{AB} f(z)\,dz + \int_{-AB} f(z)\,dz + \oint_{C_1} f(z)\,dz = 0$$

or

$$\oint_C f(z)\,dz = \oint_{C_1} f(z)\,dz. \tag{5}$$

The last result is sometimes called the **principle of deformation of contours** since we can think of the contour C_1 as a continuous deformation of the contour C. Under this deformation of contours, if the region between C and C_1 contains only points where f is analytic, then the value of the integral does not change. In other words, (5) allows us to evaluate an integral over a complicated simple closed contour C by replacing C with a contour C_1 that is more convenient.

EXAMPLE 3 Applying Deformation of Contours

Evaluate $\oint_C \dfrac{dz}{z-i}$, where C is the contour shown in black in **FIGURE 5.3.5**.

Solution Note that $f(z) = \dfrac{1}{z-i}$ is analytic on the multiply connected domain consisting of the complex plane excluding the point $z = i$. In view of (5), we choose the more convenient circular contour C_1 drawn in color in the figure. By taking the radius of the circle to be $r = 1$, we are guaranteed that C_1 lies within C. In other words, C_1 is the circle $|z - i| = 1$, which from (10) of Section 2.2 can be parametrized by $z = i + e^{it}$, $0 \le t \le 2\pi$. From $z - i = e^{it}$ and $dz = ie^{it}dt$ we obtain

$$\oint_C \frac{dz}{z-i} = \oint_{C_1} \frac{dz}{z-i} = \int_0^{2\pi} \frac{ie^{it}}{e^{it}}dt = i\int_0^{2\pi} dt = 2\pi i. \quad \equiv$$

The result obtained in Example 3 can be generalized. Using the principle of deformation of contours (5) and proceeding as in the example, it can be shown that if z_0 is any constant complex number interior to *any* simple closed contour C, then for n an integer we have

$$\oint_C \frac{dz}{(z-z_0)^n} = \begin{cases} 2\pi i, & n = 1 \\ 0, & n \neq 1. \end{cases} \tag{6}$$

For example, $1/(z-z_0)^{-3} = (z-z_0)^3$ is a polynomial. ▶

The fact that the integral in (6) is zero when $n \neq 1$ follows only partially from the Cauchy-Goursat theorem. When n is zero or a negative integer, $1/(z-z_0)^n$ is a *polynomial* and therefore entire. Theorem 5.3.1 and the discussion following Example 1 then indicate that $\oint_C dz/(z-z_0)^n = 0$. It is left as an exercise to show that the integral is still zero when n is a positive integer different from 1. See Problem 24 in Exercises 5.3.

Analyticity of the function f at all points within and on a simple closed contour C is *sufficient* to guarantee that $\oint_C f(z)\,dz = 0$. However, the result in (6) emphasizes that analyticity is not *necessary*; in other words, it can happen that $\oint_C f(z)\,dz = 0$

without f being analytic within C. For instance, if C in Example 2 is the circle $|z| = 1$, then (6), with the identifications $n = 2$ and $z_0 = 0$, immediately gives $\oint_C \frac{dz}{z^2} = 0$. Note that $f(z) = 1/z^2$ is not analytic at $z = 0$ within C.

EXAMPLE 4 Applying Formula (6)

Evaluate $\oint_C \frac{5z+7}{z^2+2z-3}dz$, where C is circle $|z-2| = 2$.

Solution Since the denominator factors as $z^2+2z-3 = (z-1)(z+3)$ the integrand fails to be analytic at $z = 1$ and $z = -3$. Of these two points, only $z = 1$ lies within the contour C, which is a circle centered at $z = 2$ of radius $r = 2$. Now by partial fractions

$$\frac{5z+7}{z^2+2z-3} = \frac{3}{z-1} + \frac{2}{z+3}$$

and so

$$\oint_C \frac{5z+7}{z^2+2z-3}\,dz = 3\oint_C \frac{1}{z-1}\,dz + 2\oint_C \frac{1}{z+3}\,dz. \tag{7}$$

In view of the result given in (6), the first integral in (7) has the value $2\pi i$, whereas the value of the second integral is 0 by the Cauchy-Goursat theorem. Hence, (7) becomes

$$\oint_C \frac{5z+7}{z^2+2z-3}\,dz = 3(2\pi i) + 2(0) = 6\pi i.$$

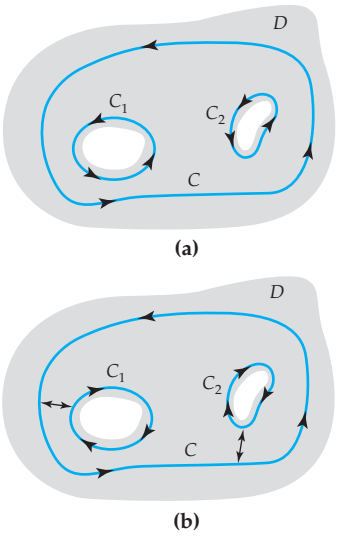


FIGURE 5.3.6 Triply connected domain D

If C, C_1, and C_2 are simple closed contours as shown in FIGURE 5.3.6(a) and if f is analytic on each of the three contours as well as at each point interior to C but exterior to both C_1 and C_2, then by introducing crosscuts between C_1 and C and between C_2 and C, as illustrated in Figure 5.3.6(b), it follows from Theorem 5.3.1 that

$$\oint_C f(z)\,dz + \oint_{C_1} f(z)\,dz + \oint_{C_2} f(z)\,dz = 0$$

and so

$$\oint_C f(z)\,dz = \oint_{C_1} f(z)\,dz + \oint_{C_2} f(z)\,dz.$$

The next theorem summarizes the general result for a multiply connected domain with n "holes."

THEOREM 5.3.2 Cauchy-Goursat Theorem for Multiply Connected Domains

Suppose C, C_1, ..., C_n are simple closed curves with a positive orientation such that C_1, C_2, ..., C_n are interior to C but the regions interior to each C_k, $k = 1, 2, \ldots, n$, have no points in common. If f is analytic on each contour and at each point interior to C but exterior to all the C_k, $k = 1, 2, \ldots, n$, then

$$\oint_C f(z)\,dz = \sum_{k=1}^{n} \oint_{C_k} f(z)\,dz. \tag{8}$$

EXAMPLE 5 Applying Theorem 5.3.2

Evaluate $\oint_C \frac{dz}{z^2+1}$, where C is the circle $|z| = 4$.

Solution In this case the denominator of the integrand factors as $z^2+1 = (z-i)(z+i)$. Consequently, the integrand $1/(z^2+1)$ is not analytic at $z = i$ and at $z = -i$. Both of these points lie within the contour C. Using partial fraction decomposition once more, we have

$$\frac{1}{z^2+1} = \frac{1}{2i}\frac{1}{z-i} - \frac{1}{2i}\frac{1}{z+i}$$

and

$$\oint_C \frac{dz}{z^2+1} = \frac{1}{2i}\oint_C \left[\frac{1}{z-i} - \frac{1}{z+i}\right] dz.$$

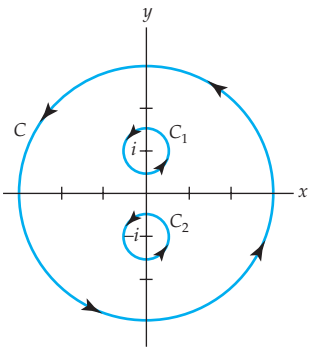


FIGURE 5.3.7 Contour for Example 5

We now surround the points $z = i$ and $z = -i$ by circular contours C_1 and C_2, respectively, that lie entirely within C. Specifically, the choice $|z-i| = \frac{1}{2}$ for C_1 and $|z+i| = \frac{1}{2}$ for C_2 will suffice. See FIGURE 5.3.7. From Theorem 5.3.2 we can write

$$\oint_C \frac{dz}{z^2+1} = \frac{1}{2i}\oint_{C_1}\left[\frac{1}{z-i} - \frac{1}{z+i}\right]dz + \frac{1}{2i}\oint_{C_2}\left[\frac{1}{z-i} - \frac{1}{z+i}\right]dz$$

$$= \frac{1}{2i}\oint_{C_1}\frac{1}{z-i}dz - \frac{1}{2i}\oint_{C_1}\frac{1}{z+i}dz + \frac{1}{2i}\oint_{C_2}\frac{1}{z-i}dz - \frac{1}{2i}\oint_{C_2}\frac{1}{z+i}dz. \quad (9)$$

Because $1/(z+i)$ is analytic on C_1 and at each point in its interior and because $1/(z-i)$ is analytic on C_2 and at each point in its interior, it follows from (4) that the second and third integrals in (9) are zero. Moreover, it follows from (6), with $n = 1$, that

$$\oint_{C_1}\frac{dz}{z-i} = 2\pi i \quad \text{and} \quad \oint_{C_2}\frac{dz}{z+i} = 2\pi i.$$

Thus (9) becomes

$$\oint_C \frac{dz}{z^2+1} = \pi - \pi = 0.$$

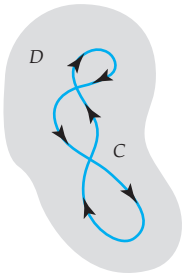


FIGURE 5.3.8 Contour C is closed but not simple.

Remarks

Throughout the foregoing discussion we assumed that C was a simple closed contour, in other words, C did not intersect itself. Although we shall not give the proof, it can be shown that the Cauchy-Goursat theorem is valid for any closed contour C in a simply connected domain D. As shown in FIGURE 5.3.8, the contour C is closed but not simple. Nevertheless, if f is analytic in D, then $\oint_C f(z)\,dz = 0$. See Problem 23 in Exercises 5.3.

EXERCISES 5.3

Answers to selected odd-numbered problems begin on page ANS-15.

In Problems 1–8, show that $\oint_C f(z)\,dz = 0$, where f is the given function and C is the unit circle $|z| = 1$.

1. $f(z) = z^3 - 1 + 3i$
2. $f(z) = z^2 + \dfrac{1}{z-4}$
3. $f(z) = \dfrac{z}{2z+3}$
4. $f(z) = \dfrac{z-3}{z^2+2z+2}$
5. $f(z) = \dfrac{\sin z}{(z^2-25)(z^2+9)}$
6. $f(z) = \dfrac{e^z}{2z^2+11z+15}$
7. $f(z) = \tan z$
8. $f(z) = \dfrac{z^2-9}{\cosh z}$

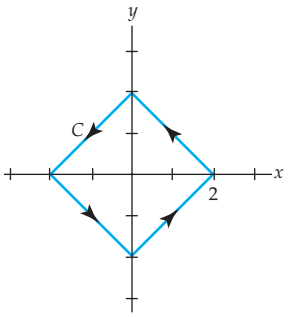


FIGURE 5.3.9 Figure for Problem 9

9. Evaluate $\oint_C \dfrac{1}{z}\,dz$, where C is the contour shown in FIGURE 5.3.9.

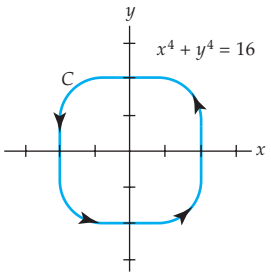


FIGURE 5.3.10 Figure for Problem 10

10. Evaluate $\oint_C \dfrac{5}{z+1+i}\,dz$, where C is the contour shown in FIGURE 5.3.10.

In Problems 11–22, use any of the results in this section to evaluate the given integral along the indicated closed contour(s).

11. $\oint_C \left(z + \dfrac{1}{z}\right) dz;\ |z| = 2$
12. $\oint_C \left(z + \dfrac{1}{z^2}\right) dz;\ |z| = 2$
13. $\oint_C \dfrac{z}{z^2-\pi^2}\,dz;\ |z| = 3$
14. $\oint_C \dfrac{10}{(z+i)^4}\,dz;\ |z+i| = 1$
15. $\oint_C \dfrac{2z+1}{z^2+z}\,dz;$ **(a)** $|z| = \frac{1}{2}$, **(b)** $|z| = 2$, **(c)** $|z-3i| = 1$
16. $\oint_C \dfrac{2z}{z^2+3}\,dz;$ **(a)** $|z| = 1$, **(b)** $|z-2i| = 1$, **(c)** $|z| = 4$
17. $\oint_C \dfrac{-3z+2}{z^2-8z+12}\,dz;$ **(a)** $|z-5| = 2$, **(b)** $|z| = 9$
18. $\oint_C \left(\dfrac{3}{z+2} - \dfrac{1}{z-2i}\right) dz;$ **(a)** $|z| = 5$, **(b)** $|z-2i| = \frac{1}{2}$
19. $\oint_C \dfrac{z-1}{z(z-i)(z-3i)}\,dz;\ |z-i| = \frac{1}{2}$
20. $\oint_C \dfrac{1}{z^3+2iz^2}\,dz;\ |z| = 1$
21. $\oint_C \operatorname{Ln}(z+10)\,dz;\ |z| = 2$
22. $\oint_C \left[\dfrac{5}{(z-2)^3} + \dfrac{3}{(z-2)^2} - \dfrac{10}{z-2} + 7\csc z\right] dz;\ |z-2| = \frac{1}{2}$

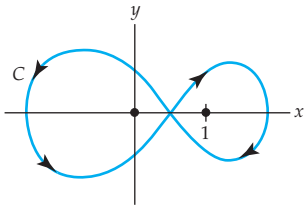


FIGURE 5.3.11 Figure for Problem 23

23. Evaluate $\oint_C \dfrac{8z-3}{z^2-z}\,dz$, where C is the "figure-eight" contour shown in FIGURE 5.3.11. [*Hint*: Express C as the union of two closed curves C_1 and C_2.]
24. Suppose z_0 is any constant complex number interior to any simple closed curve contour C. Show that for a positive integer n,

$$\oint_C \frac{dz}{(z-z_0)^n} = \begin{cases} 2\pi i, & n = 1 \\ 0, & n > 1. \end{cases}$$

In Problems 25 and 26, evaluate the given contour integral by any means.

25. $\oint_C \left(\dfrac{e^z}{z+3} - 3\bar{z}\right) dz$, where C is the unit circle $|z| = 1$

26. $\oint_C \left(z^3 + z^2 + \text{Re}(z)\right) dz$, where C is the triangle with vertices $z = 0$, $z = 1 + 2i$, and $z = 1$

Focus on Concepts

27. Explain why $\oint_C f(z)\,dz = 0$ for each of the following functions and C is any simple closed contour in the complex plane.

(a) $f(z) = (5iz^4 - 4z^2 + 2 - 6i)^9$ **(b)** $f(z) = (z^2 - 3iz)e^{5z}$

(c) $f(z) = \dfrac{\sin z}{e^{z^2}}$ **(d)** $f(z) = z\cos^2 z$

28. Describe contours C for which we are guaranteed that $\oint_C f(z)\,dz = 0$ for each of the following functions.

(a) $f(z) = \dfrac{1}{z^3 + z}$ **(b)** $f(z) = \csc z$

(c) $f(z) = \dfrac{1}{1 - e^z}$ **(d)** $f(z) = \text{Ln}\, z$

29. Explain why the integral in Problem 25 is the same as

$$\oint_C \left(\frac{e^z}{z+3} - \frac{3}{z}\right) dz$$

and why, in view of (6), this form makes the integral slightly easier to evaluate.

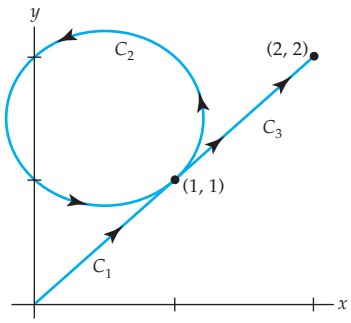


FIGURE 5.3.12 Figure for Problem 30

30. Evaluate $\int_C e^z\,dz$ from $z = 0$ to $z = 2 + 2i$ on the contour C shown in **FIGURE 5.3.12** that consists of the line $y = x$ and a circle tangent to the line at (1, 1).

31. From Example 1 we know the value of $\oint_C e^z dz$ for any simple closed contour C in the complex plane. In particular, use $|z| = 1$ as C and the parametrization $z = e^{i\theta}$, $0 \le \theta \le 2\pi$, to discover the values of the real integrals

$$\int_0^{2\pi} e^{\cos\theta}\sin(\theta + \sin\theta)d\theta \quad \text{and} \quad \int_0^{2\pi} e^{\cos\theta}\cos(\theta + \sin\theta)d\theta.$$

32. Let n be a positive integer. Use a complex contour integral to show that

$$\int_0^{2\pi} e^{\sin n\theta}\cos(\theta - \cos n\theta)\,d\theta = 0 \quad \text{and} \quad \int_0^{2\pi} e^{\sin n\theta}\sin(\theta - \cos n\theta)\,d\theta = 0.$$

[*Hint*: Modify the integrand from Problem 31.]

33. Which of the domains shown in Figure 1.5.5 on page 29 are simply connected?

34. Suppose D_1 and D_2 are simply connected domains in the complex plane.

(a) Is the union $D_1 \cup D_2$ a simply connected domain? If you think the statement is true, try to prove it. If you think the statement is false, then give a counterexample.

(b) Repeat part (a) for the intersection $D_1 \cap D_2$.